

PROBLEM SHEET 4

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The following questions are about the simply-typed λ -calculus (STLC).

1. (*) Prove progress and preservation for the **constants-and-functions fragment** of the STLC.

The constants-and-function fragment of the STLC is an extension to the language of numbers and strings: we reached it by *adding* the rules for function types. Thus, to establish these theorems **you only need to show them for the new function rules**, as last week's proofs cover the rest! (We will ignore sums and products in this question!)

Do this in steps:

1. Extend the key lemmata (you may assume weakening, but you can also prove it if you feel like it):
 - (a) Inversion
 - (b) Substitution
 - (c) Canonical forms
 2. Prove preservation
 3. Prove progress
2. (*) Do you need to prove progress and preservation for the CBV STLC, or can you say that they have already been established? Do not do any proofs in the answer to this question (unless you want to).